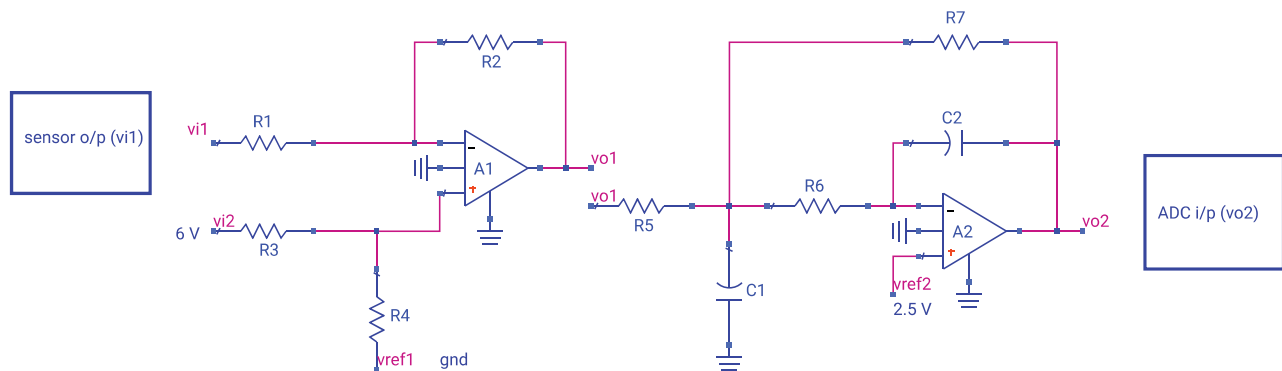


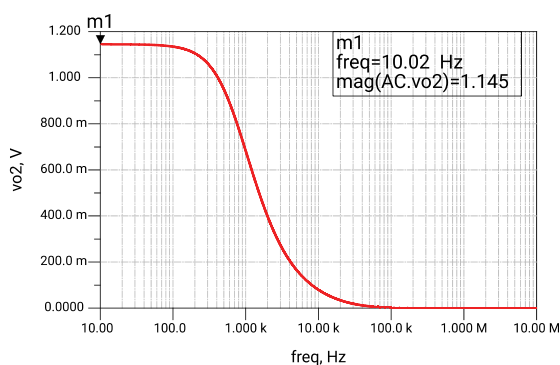
Measurement of height using a sensor module

OpAmp circuit diagram:



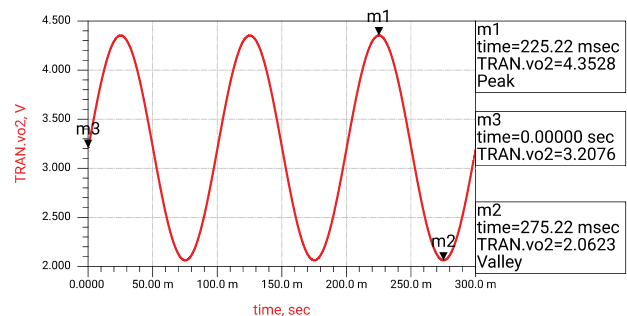
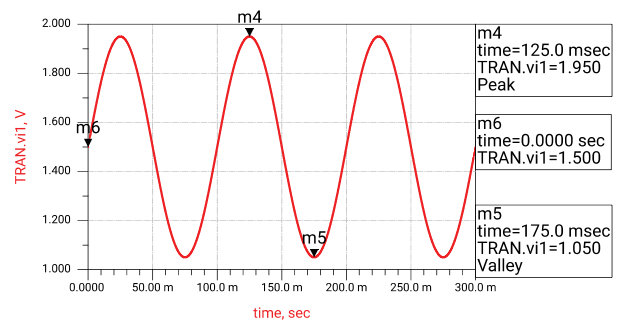
Note: here sensor o/p voltage is between [1.05, 1.95] V & o/p voltage vo2 is between [0, 5] V to fed ADC.

AC Analysis:

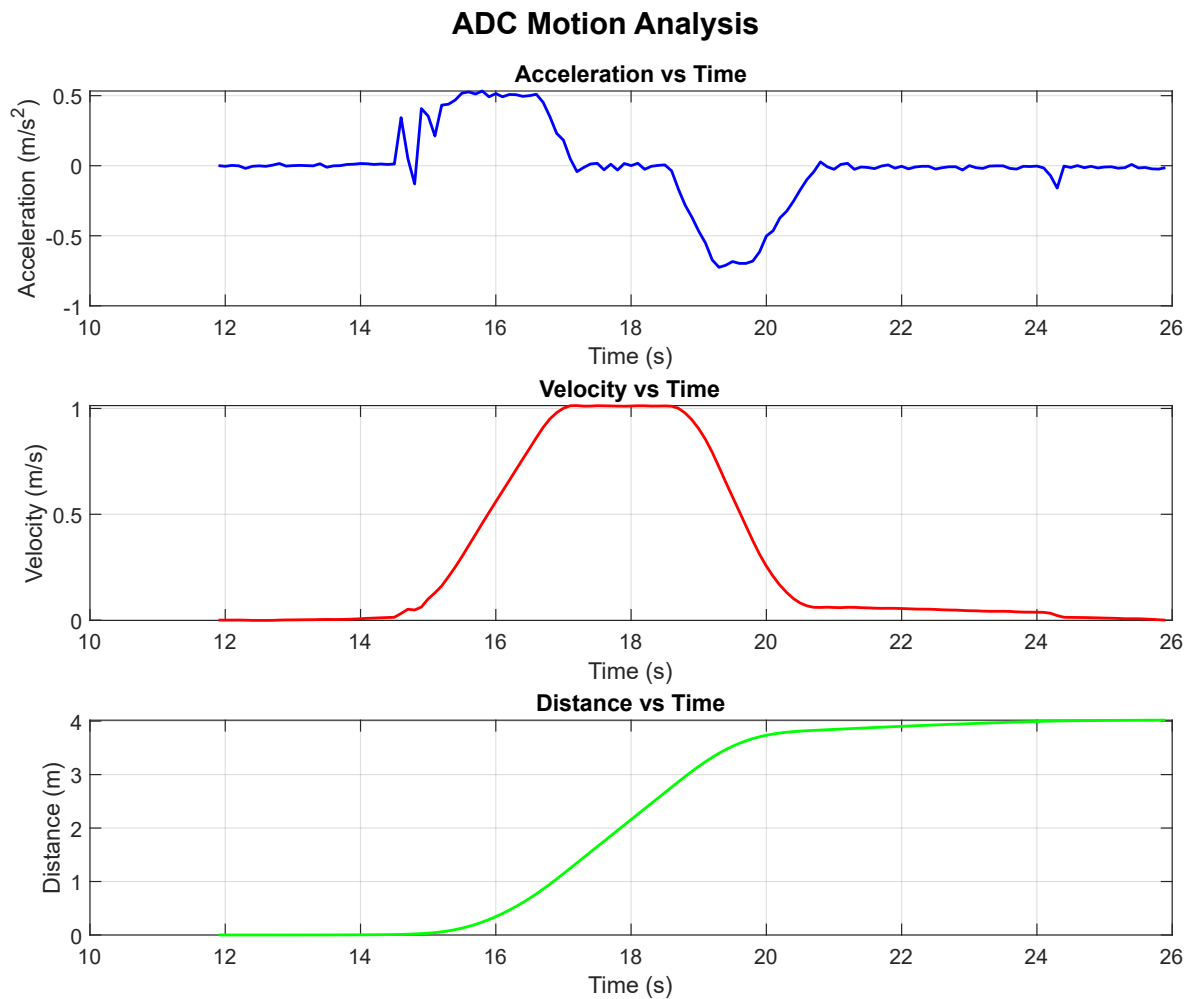


DC.vo2	u_vo2	d_vo2
3.2076 V	4.3528	2.0623

Transient Analysis:



After converting ADC data to acceleration & calibrating it:



References:

- [1] R. C. Jaeger and T. N. Blalock, Microelectronic Circuit Design, 4th ed.
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- [3] R. A. Gayakwad, Op-Amps and Linear Integrated Circuits, 4th ed.
Upper Saddle River, NJ, USA: Prentice Hall, 2000.